


BIG IDEAS:

- A **unit fraction** is a positive fraction with a numerator of 1
- Multiplying or dividing a fraction's numerator and denominator by the same number (except 0) does not change its value
- **Common denominators or numerators** are helpful for comparing fractions

EXAMPLE 1

$$\text{a) } \frac{1}{3} = \frac{5}{15}$$

$\times 5$
 $\div 5$

$$\text{b) } -\frac{6}{12} = -\frac{1}{2}$$

$\div 6$
 $\div 6$

EXAMPLE 2

$$\text{a) } \frac{22}{35} > \frac{19}{35}$$

common denominator

$$\text{b) } \frac{3}{7} < \frac{3}{5}$$

common numerator

EXAMPLE 3

$$\frac{13}{5} = 2\frac{3}{5}$$

5 goes into 13 two full times, with a remainder of 3.

EXAMPLE 4

$$\begin{aligned} 2\frac{3}{4} &= 2 + \frac{3}{4} \\ &= 1 + 1 + \frac{3}{4} \\ &= \frac{4}{4} + \frac{4}{4} + \frac{3}{4} \\ &= \frac{11}{4} \end{aligned}$$

$2 \times 4 + 3 = 11$

MORE EXAMPLES

Things to remember...


THINKING BEYOND...

1. Represent the fraction $\frac{5}{8}$ as a decimal and as a percentage.
2. When holding 50 gallons of water, a tank is $\frac{2}{5}$ full. How many gallons will the entire tank hold?
3. Which of the **continued fractions** shown on the right is greater? How do you know?
4. Express 0.321321321321... as a fraction.

$$1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

$$\text{or } 2 + \frac{2}{2 + \frac{2}{2 + \frac{2}{2 + \dots}}}$$